

Course Name: Advanced System Analysis and
Software Design

Course Code: ENSF 480

Assignment Number: Lab 4

Lab Section: B01

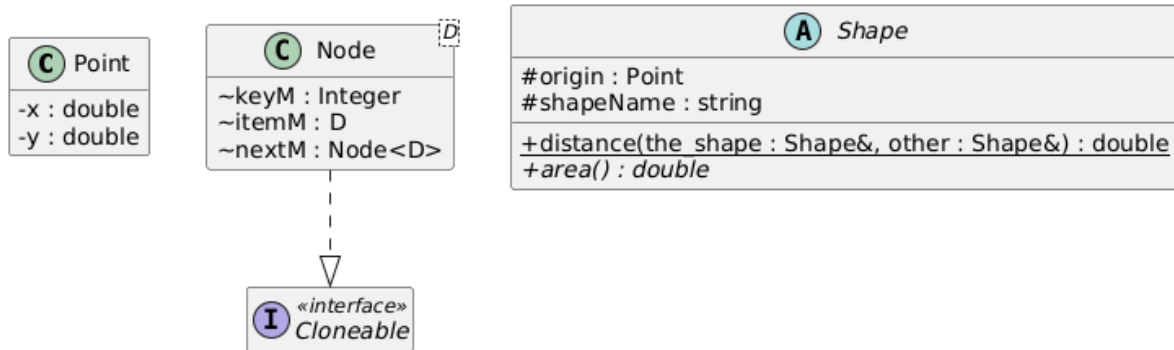
Completed by:

Aayush Karthikeyan (UCID# 30189743)

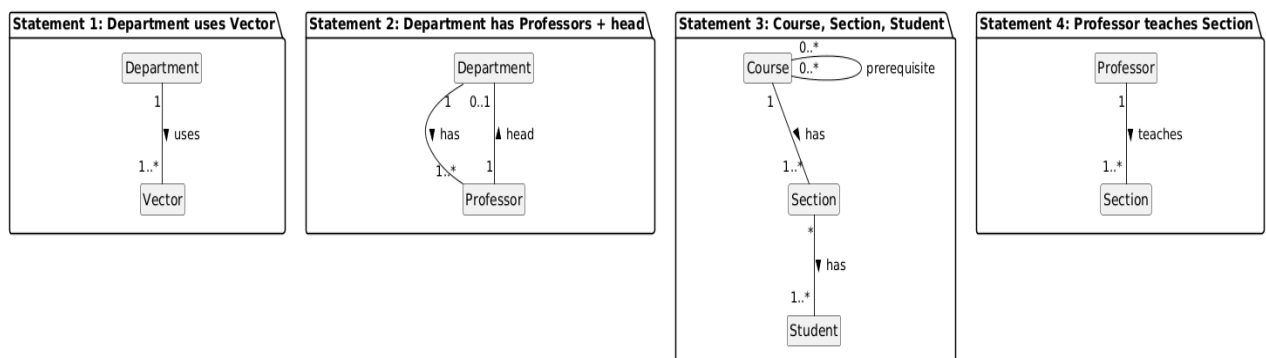
Sarvesh Vettrivelan (UCID# 30242015)

Submission Date: 04/10/2026

Exercise A - UML Class Notation



Exercise B - Association and Multiplicity



Exercise C - From Diagram to Code

```
/*
 * File Name: lab4exe_C.h
 * Assignment: Lab 4 Exercise C
 * Completed By:
 *     - Aayush Karthikeyan (UCID# 30189743)
 *     - Sarvesh Vettrivelan (UCID# 30242015)
 * Submission Date: Oct. 4, 2026
 */

#ifndef LAB4EXE_C_H
#define LAB4EXE_C_H

#include <string>
using std::string;

// «interface» Moveable
class Moveable {
public:
    virtual void forward() = 0;
    virtual void backward() = 0;
    virtual ~Moveable() {}
};

// Resizable (lollipop notation = interface)
class Resizable {
public:
    virtual void enlarge(int n) = 0;
    virtual void shrink(int n) = 0;
    virtual ~Resizable() {}
};
```

```
// Abstract class (italic name) realizing both interfaces
class Vehicle : public Moveable, public Resizable {
protected:
    string name;
public:
    Vehicle(string name);
    virtual void move() = 0;
    virtual ~Vehicle() {}
};

// {leaf} -> final: cannot be inherited from
class Car final : public Vehicle {
private:
    int seats;
public:
    void turn();
};

#endif
```

Exercise D – Reverse Engineering a Java Program

